COLLEGE OF ENGINEERING VIZIANAGARAM

Department of Information Technology

QUANTUM COMPUTING

LECTURE NOTES

UNIT V: Quantum Computing Applications & Emerging Trends

Course Structure: 300 Minutes / 5 Hours

Unit V — Syllabus Coverage

This unit connects the quantum-computing foundations, circuits, algorithms, and programming concepts from the earlier units to application domains. It introduces quantum cryptography and key distribution, post-quantum cryptography, quantum machine learning, optimization and quantum annealing, quantum chemistry and materials simulation, and the quantum-computing industry and career landscape. The unit closes with project guidance and directions for further study.

Topics Covered

- Introduction to Unit V
- Quantum Cryptography and Key Distribution
- Post-Quantum Cryptography
- Quantum Machine Learning
- Optimization and Quantum Annealing
- Quantum Chemistry and Materials Simulation
- Industry Landscape and Career Pathways
- Course Wrap-Up and Project Guidance

Unit Learning Outcomes

1. Explain how quantum-computing concepts are reused across application domains through the pipeline encode → evolve → measure → post-process.
2. Describe the physical principles behind quantum key distribution and explain the BB84 and E91 protocols.
3. Explain the quantum threat to RSA, Diffie–Hellman and ECC and distinguish post-quantum cryptography from quantum key distribution.
4. Describe common approaches to quantum machine learning, including data encoding, parameterized circuits, VQE and quantum kernels.
5. Formulate optimization problems using QUBO/Ising ideas and explain the basic principles of quantum annealing and QAOA.
6. Explain how quantum computing can be applied to molecular and materials simulation using mappings such as Jordan–Wigner and Bravyi–Kitaev.
7. Identify major roles, skills and learning pathways in the quantum-computing ecosystem.
8. Design and document a small quantum-computing project using the concepts covered in the course.

Part A: Introduction to Quantum Computing Applications

A.1 Common Application Pipeline

The application topics in this unit reuse a common computational pattern: encode → evolve → measure → post-process. Classical information is first represented in a form suitable for a quantum circuit or quantum system. Quantum operations transform that representation. Measurement produces classical information, and classical post-processing converts the measured data into the required application result.

Stage	Meaning	Example
Encode	Represent the problem or data in a quantum state	Encode a search item or molecular Hamiltonian
Evolve	Apply quantum gates or physical evolution	Run a QAOA circuit
Measure	Convert quantum information into classical outcomes	Measure computational-basis bit strings
Post-process	Interpret and analyze the measured results	Estimate a cost, classify data or identify a key

A.2 NISQ vs Fault-Tolerant Applications

Aspect	Near-Term / NISQ	Fault-Tolerant Era
Hardware	Noisy and resource constrained	Error-corrected logical computation
Typical concern	Noise, circuit depth and sampling overhead	Logical reliability and large-scale algorithms
Application focus	Experiments, hybrid algorithms and selected demonstrations	Larger algorithms requiring reliable logical qubits
Examples in this unit	Variational circuits, QAOA, small simulations	Large-scale cryptographic and simulation applications

Part B: Quantum Cryptography and Key Distribution

B.1 Classical Key Exchange and the Quantum Problem

Cryptographic communication requires Alice and Bob to establish or use secret information while preventing an adversary from obtaining it unnoticed. Quantum key distribution uses properties of quantum systems to detect disturbance caused by an eavesdropper during key establishment.

B.2 Physical Principles Behind QKD

- No-cloning: an unknown quantum state cannot be copied perfectly.
- Measurement disturbance: measuring a quantum state in an incompatible basis can change the state.
- Statistical detection: Alice and Bob can compare selected results to estimate whether the channel has been disturbed.

B.3 BB84 Protocol

BB84 is a prepare-and-measure quantum key-distribution protocol. Alice prepares quantum states using two possible bases, while Bob independently chooses measurement bases. After transmission, they publicly compare bases but do not reveal the corresponding key values for the retained positions.

BB84 Step-by-Step

9. Alice randomly chooses a bit string.
10. Alice randomly chooses a basis for each bit.

11. Alice prepares a quantum state according to the bit and basis choices.
12. Alice sends the quantum states to Bob through the quantum channel.
13. Bob randomly chooses a measurement basis for each received state.
14. Alice and Bob publicly compare their basis choices.
15. They keep positions where their bases agree. This is called sifting.
16. They reveal a selected subset of retained bits to estimate the error rate.
17. If the estimated error is acceptable, they continue with post-processing such as error correction and privacy amplification.
18. The remaining information forms the shared secret key.

B.4 Intercept-and-Resend Attack

In an intercept-and-resend attack, an eavesdropper, Eve, measures transmitted quantum states and then sends replacement states to Bob. When Eve uses the wrong basis, her measurement can disturb the state. The disturbance creates errors in the subset where Alice and Bob later compare results.

The course outline uses the 25% error signature as the characteristic idealized error rate associated with intercept-and-resend when the relevant basis choices are random and the standard BB84 assumptions are used.

B.5 E91 Protocol

E91 is an entanglement-based quantum key-distribution protocol. Instead of Alice independently preparing individual states for transmission, entangled pairs are used and measurements performed by separated parties are analyzed for the correlations associated with entanglement.

B.6 Practical QKD

- Fibre links can carry quantum states over terrestrial distances.
- Satellites can be used for long-distance quantum communication scenarios.
- Trusted nodes can form intermediate points in a larger network architecture.
- Real deployments require consideration of hardware, channel loss, detector behavior, synchronization and system security.

Worked Example 1: BB84 Basis Sifting

Position	Alice Basis	Bob Basis	Keep?
1	Z	Z	Yes
2	Z	X	No
3	X	X	Yes
4	X	Z	No
5	Z	Z	Yes
6	X	X	Yes

Only positions 1, 3, 5 and 6 are retained after basis sifting. The actual bit values are not shown during this public basis-comparison step.

Worked Example 2: Intercept-and-Resend Error Probability

Under the standard simplified BB84 analysis, Alice and Bob use the same basis with probability 1/2. When Eve chooses the wrong basis, which occurs with probability 1/2, her resend causes Bob to obtain the wrong bit with probability 1/2 when Bob uses the original basis.

Therefore the error probability in the sifted key is approximately $1/2 \times 1/2 = 1/4 = 25\%$ for the idealized intercept-and-resend attack.

Part C: Post-Quantum Cryptography

C.1 What Is Post-Quantum Cryptography?

Post-quantum cryptography (PQC) refers to classical cryptographic algorithms designed to remain secure against adversaries equipped with sufficiently capable quantum computers. PQC is different from QKD: PQC uses classical computational cryptography, while QKD uses quantum communication to establish keys.

C.2 The Shor Threat Model

Shor's algorithm provides a quantum speedup for integer factoring and discrete logarithm problems. Therefore, public-key systems whose security relies on these mathematical problems are relevant to the quantum-threat model.

Classical Primitive	Quantum Threat Discussed in This Course
RSA	Factoring-based public-key security is threatened by Shor's algorithm
Diffie–Hellman	Discrete-logarithm-based security is threatened by Shor's algorithm
ECC	Elliptic-curve discrete-logarithm security is threatened by Shor's algorithm
Symmetric cryptography	Grover gives a quadratic search speedup; larger key sizes can address the generic search reduction
Cryptographic hashes	Grover can reduce generic preimage-search security; longer output sizes can provide additional margin

C.3 Harvest Now, Decrypt Later

The phrase describes an information-security concern in which encrypted information is collected today and retained with the expectation that future technology may make decryption easier. This creates a motivation for migration toward quantum-resistant cryptographic mechanisms before large-scale cryptographically relevant quantum computers exist.

C.4 PQC Families in the Course Outline

- Lattice-based cryptography: represented in the outline by Kyber and Dilithium.
- Hash-based cryptography.
- Code-based cryptography.
- Multivariate cryptography.

C.5 Standardization and Migration

The course outline includes NIST post-quantum cryptography standardization status and timeline as a topic. It also identifies crypto-agility, certificates and embedded devices as migration challenges.

Crypto-agility means that systems can replace cryptographic algorithms without redesigning the entire application or infrastructure. This is important because large organizations may have long-lived certificates, protocols, devices and software dependencies.

Worked Example 3: Why Symmetric Keys Need Different Treatment

Suppose a generic quantum search method provides a quadratic speedup over exhaustive search. Conceptually, a search space of size 2^k changes from a classical scale proportional to 2^k to a quantum scale proportional to 2^(k/2). This is why increasing a symmetric key length can compensate for the

generic quadratic reduction, whereas RSA/ECC require a different response because Shor's algorithm attacks their underlying mathematical structures.

Part D: Quantum Machine Learning

D.1 Where Quantum Computing Could Help ML

- Data encoding: representing classical data in quantum states.
- Quantum kernels: using quantum circuits to construct feature-space relationships.
- Optimization: using hybrid quantum-classical procedures for parameter optimization.
- Parameterized quantum circuits: using trainable quantum gates as model components.

D.2 Classical Data Encoding

Encoding	Basic Idea
Basis encoding	Represent information using computational-basis states
Amplitude encoding	Store normalized data values in amplitudes of a quantum state
Angle encoding	Map classical numerical values to rotation angles

D.3 Parameterized Quantum Circuits

A parameterized quantum circuit contains gates whose rotation angles or other parameters are controlled by adjustable classical values. The circuit is executed, measurements are collected, and a classical optimizer updates the parameters.

D.4 Variational Loop

The basic hybrid loop is: ansatz → measure → classical optimizer → update parameters → repeat.

- Ansatz: choose a parameterized circuit structure.
- Measure: execute the circuit and obtain an objective or expectation estimate.
- Classical optimizer: compare the result with the target objective and propose new parameters.
- Iteration: repeat until the selected stopping condition is reached.

D.5 Variational Quantum Eigensolver (VQE)

VQE is a hybrid variational method for estimating the ground-state energy of a quantum system. A parameterized state is prepared, measurements estimate the expectation value of the Hamiltonian, and a classical optimizer adjusts the circuit parameters to reduce the estimated energy.

D.6 Quantum Kernel Methods and Quantum SVMs

A quantum kernel method uses a quantum circuit to map classical data into a quantum feature space and estimate relationships between data points. These kernel values can then be supplied to a classical machine-learning method such as a support vector machine.

D.7 Variational Quantum Classifier

A variational quantum classifier uses encoded input data and trainable circuit parameters to produce a measurement-based classification output. Training follows the hybrid optimization loop.

D.8 Barren Plateaus and Training Obstacles

Barren plateaus describe situations in which gradients of a parameterized quantum circuit become extremely small, making optimization difficult. Other challenges include noise, limited circuit depth, measurement sampling, data encoding costs and optimization instability.

D.9 Benchmarking Quantum and Classical ML

A meaningful comparison should use the same dataset, clearly defined metrics and comparable experimental conditions. Small demonstrations should not automatically be interpreted as evidence of a general quantum advantage; the course outline explicitly includes benchmarking and examination of what the evidence shows.

Worked Example 4: Simple Variational Optimization

Suppose a one-qubit circuit prepares $|\psi(\theta)\rangle = Ry(\theta)|0\rangle$. A classical optimizer chooses θ, the circuit is measured, and an objective value is calculated from the measurement results. The optimizer then changes θ and repeats the experiment. This simple example illustrates the structure of a hybrid quantum-classical variational algorithm.

Part E: Optimization and Quantum Annealing

E.1 Combinatorial Optimization

Many practical problems require selecting the best combination from a large set of possibilities. Examples in the course outline include portfolio selection, logistics, routing and supply-chain scheduling.

E.2 QUBO Formulation

A Quadratic Unconstrained Binary Optimization (QUBO) problem uses binary variables $x_i \in \{0,1\}$ and an objective of the form:

$$Q(x) = \Sigma_i a_i x_i + \Sigma_{i<j} b_{ij} x_i x_j.$$

The goal is to find the binary assignment that minimizes or maximizes the chosen objective, depending on the formulation.

E.3 Ising Formulation

An Ising model commonly uses spin variables $s_i \in \{-1,+1\}$. A quadratic energy function can be written as $E(s) = \Sigma_i h_i s_i + \Sigma_{i<j} J_{ij} s_i s_j$. Binary optimization problems can be transformed into equivalent Ising forms through a variable transformation.

E.4 Quantum Annealing

Quantum annealing is an approach to optimization in which a physical quantum system is guided from an initial state toward a low-energy configuration of a problem Hamiltonian. The final measured configuration is interpreted as a candidate solution to the optimization problem.

E.5 D-Wave Hardware

The course outline introduces D-Wave hardware, its architecture, embedding process and limitations. Embedding maps the logical problem graph onto the physical connectivity available in the hardware. This can require multiple physical resources to represent a logical variable or interaction.

E.6 QAOA

The Quantum Approximate Optimization Algorithm is a gate-based variational approach for combinatorial optimization. Its circuit alternates between a cost layer, which encodes the objective, and a mixer layer, which explores candidate states.

19. Prepare an initial superposition.
20. Apply a cost Hamiltonian layer.
21. Apply a mixer layer.
22. Repeat the layers for the selected depth p.
23. Measure candidate solutions.
24. Use a classical optimizer to update the circuit parameters.

E.7 Max-Cut Worked Example

Consider a graph with vertices A and B connected by one edge. Max-Cut asks for a partition that places the two connected vertices in different groups. Assign binary variables x_A and x_B. The cut is active when x_A ≠ x_B.

Assignment	Different groups?	Cut Value
00	No	0
01	Yes	1
10	Yes	1
11	No	0

Thus 01 and 10 are optimal assignments for this one-edge Max-Cut example. A QAOA circuit encodes the cost and mixer operations so that measurement produces candidate bit strings.

E.8 Application Domains

- Portfolio optimization and risk analysis in finance.
- Vehicle routing and logistics.
- Supply-chain scheduling.
- Other combinatorial problems that can be represented as binary optimization models.

Part F: Quantum Chemistry and Materials Simulation

F.1 Why Simulate Quantum Systems?

The course outline connects quantum computing to Feynman's motivation of using quantum systems to simulate other quantum systems. Molecular and materials behavior is fundamentally quantum mechanical, while exact classical simulation can become difficult as system size increases.

F.2 Electronic Structure

Electronic-structure problems seek information such as the energy and state of electrons in molecules or materials. The number of interacting degrees of freedom can make classical computation expensive, motivating quantum simulation approaches.

F.3 Mapping Fermions to Qubits

Mapping	Purpose
Jordan–Wigner	Maps fermionic operators to qubit operators while preserving the required anticommutation structure
Bravyi–Kitaev	Provides an alternative fermion-to-qubit transformation with a different representation of

	occupation and parity information

F.4 Ground-State Energy of H_2 with VQE

A typical workflow for a small molecular example is:

25. Define the molecular system and electronic Hamiltonian.
26. Transform the fermionic Hamiltonian into a qubit Hamiltonian.
27. Choose a parameterized quantum circuit.
28. Measure the expectation values of the required Hamiltonian terms.
29. Combine the measured terms to estimate the total energy.
30. Use a classical optimizer to adjust circuit parameters and reduce the energy estimate.

F.5 Applications

- Drug discovery and molecular design.
- Battery research.
- Catalyst research.
- Materials research.

F.6 Realistic Assessment

The unit distinguishes between what is demonstrated and what is projected. Small quantum chemistry demonstrations can be implemented on present systems, while larger and more accurate simulations may require improvements in qubit quality, error correction, circuit resources and scalability.

Worked Example 5: Hamiltonian Expectation Value

Suppose a simplified qubit Hamiltonian is $H = 0.5Z + 0.2X$ and a circuit produces estimates $\langle Z \rangle = -0.8$ and $\langle X \rangle = 0.3$. The estimated energy is:

$$E = 0.5\langle Z \rangle + 0.2\langle X \rangle = 0.5(-0.8) + 0.2(0.3) = -0.40 + 0.06 = -0.34.$$

A variational algorithm would attempt to adjust its parameters to reduce this energy estimate toward the lowest accessible value.

Part G: Industry Landscape and Career Pathways

G.1 Hardware Landscape

The course outline identifies the following hardware players as examples of the quantum-computing ecosystem:

- IBM
- Google
- IonQ
- Quantinuum
- Rigetti
- PsiQuantum
- QuEra

G.2 National Missions and Funding

The outline includes national missions and funding, including India's National Quantum Mission, as part of the industry landscape. This topic connects research, infrastructure development, workforce development and the broader growth of quantum technologies.

G.3 Hardware Roadmaps

Quantum hardware roadmaps describe progression toward increasingly capable systems and ultimately fault-tolerant quantum computing. Important themes include improving qubit quality, scaling system size, reducing errors, increasing connectivity or effective connectivity, and developing error-correction capabilities.

G.4 Quantum-Computing Roles

Role	Typical Focus
Quantum Software Engineer	Quantum SDKs, circuits, compilation, simulation and software integration
Algorithm Researcher	Design and analysis of quantum algorithms and complexity
Hardware Engineer	Qubit technologies, control systems, fabrication and hardware characterization
Applications Scientist	Applying quantum methods to domain-specific problems

G.5 Skills and Portfolio

- Quantum-computing fundamentals and linear algebra.
- Quantum programming and SDK experience.
- Classical programming and data analysis.
- Knowledge of algorithms, optimization or domain-specific applications.
- Certifications and structured courses.
- Open-source contributions.
- Research internships and competitions.
- Small, well-documented quantum projects.

G.6 Further Study

- Quantum information theory.
- Quantum error correction.
- Post-quantum cryptography.
- MOOCs and textbooks.
- Quantum-computing communities.
- Competitions and practical simulation exercises.

Part H: Course Wrap-Up and Project Guidance

H.1 Course-Wide Connection

Unit	Main Contribution
Unit I	Quantum states, superposition, measurement and entanglement
Unit II	Quantum gates, circuits and universal computation
Unit III	Quantum algorithms such as Deutsch–Jozsa, QFT, Grover and Shor

Unit IV	Programming, simulation, hardware, noise and error correction
Unit V	Applications, industry context and project development

H.2 Suggested Mini-Projects

- BB84 simulator with eavesdropper detection.
- Grover demonstration for a search space of 4–8 elements.
- Variational quantum classifier.
- Quantum random-number generator.
- Classical vs quantum factoring comparison.

H.3 Project Report Structure

31. Title and problem statement.
32. Background and motivation.
33. Objective.
34. Quantum approach and algorithm.
35. Circuit or system design.
36. Implementation and tools.
37. Experimental setup.
38. Results and observations.
39. Limitations.
40. Conclusion and future work.
41. References.

Worked Example 6: Choosing a Mini-Project

Suppose a student wants a project with a small circuit and an easily observable output. A Grover demonstration for 4–8 elements can be structured as follows: define a small search problem, construct an oracle, apply Grover iterations, measure the circuit, repeat the experiment for multiple shots, and compare the measured distribution with the expected marked state. The project can then document circuit design, simulation results and limitations.

Unit Summary

- Quantum applications reuse the general pipeline encode → evolve → measure → post-process.
- QKD uses quantum properties such as no-cloning and measurement disturbance to support detection of eavesdropping.
- BB84 uses two measurement bases, basis sifting, error estimation and privacy amplification.
- The idealized intercept-and-resend attack in BB84 produces a 25% error signature under the standard random-basis assumptions.
- E91 is an entanglement-based QKD approach.
- Post-quantum cryptography uses classical cryptographic techniques designed to resist quantum attacks.
- Shor's algorithm threatens RSA, Diffie–Hellman and ECC through its relevance to factoring and discrete logarithms.
- Grover's quadratic search speedup motivates longer parameters for symmetric cryptography and hashes rather than the same response used for public-key systems.
- Quantum machine learning includes data encoding, parameterized circuits, VQE, quantum kernels and hybrid optimization.
- Barren plateaus, noise and data-encoding costs are among the training challenges discussed.

- QUBO and Ising formulations represent combinatorial optimization problems in mathematical forms suitable for quantum optimization methods.
- Quantum annealing and QAOA provide different approaches to optimization.
- Quantum chemistry uses qubit mappings such as Jordan–Wigner and Bravyi–Kitaev to represent fermionic problems.
- VQE can be used to estimate ground-state energies in small molecular examples.
- The quantum industry includes hardware, software, algorithms and applications roles, supported by skills development, projects and research experience.
- Mini-projects provide a practical bridge from course concepts to implementation.

Review Questions

42. What is the common encode → evolve → measure → post-process pipeline?
43. Differentiate NISQ-era and fault-tolerant-era applications.
44. What physical principles make QKD possible?
45. Explain the BB84 protocol step by step.
46. What is basis sifting in BB84?
47. Explain the intercept-and-resend attack and the 25% error signature.
48. What is the basic idea of E91?
49. Differentiate QKD and post-quantum cryptography.
50. Why are RSA, Diffie–Hellman and ECC relevant to the quantum threat model?
51. Explain the idea of harvest now, decrypt later.
52. Why can symmetric cryptographic security be strengthened by increasing key length in response to Grover's quadratic speedup?
53. List the PQC families included in this unit.
54. What is crypto-agility and why is it important for migration?
55. What are basis, amplitude and angle encoding?
56. Explain the variational quantum-classical loop.
57. What is VQE?
58. What are quantum kernel methods?
59. What is a barren plateau?
60. Why should quantum ML be benchmarked against classical ML?
61. What is a QUBO formulation?
62. Differentiate QUBO and Ising variables.
63. Explain the basic principle of quantum annealing.
64. What is embedding in quantum annealing hardware?
65. Explain the cost and mixer layers of QAOA.
66. Describe the Max-Cut problem with a simple graph.
67. Why is quantum simulation relevant to chemistry and materials?
68. What are Jordan–Wigner and Bravyi–Kitaev transformations?
69. Explain the workflow for estimating H_2 ground-state energy using VQE.
70. List the quantum-computing roles discussed in the unit.
71. What skills can be included in a quantum-computing portfolio?
72. List the suggested mini-projects for the course.
73. How should a quantum-computing project report be structured?

Suggested Practical Activities

- Simulate a complete BB84 exchange and introduce an intercept-and-resend attacker.

- Implement basis sifting and calculate the observed error rate.
- Build a small variational classifier on a toy dataset.
- Formulate a two- or three-variable optimization problem as QUBO.
- Implement a small QAOA Max-Cut example.
- Calculate an expectation value for a simple qubit Hamiltonian.
- Create a small VQE workflow for a toy Hamiltonian.
- Prepare a mini-project report using the recommended structure.